EDS 223: Homework Assignment 3

United States West Coast Aquaculture Potential Analysis

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```
# Load in packages
library(tidyverse)
library(spDataLarge) # R package for large datasets for spatial analysis
library(sf) # spatial vector data
library(stars) # raster data
library(terra) # raster data
library(tmap) # map
library(viridisLite)
library(kableExtra)
```

Load in and clean data

```
# Load in data

## shapefile for the West Coast EEZ
west_cost_eez <- st_read('data/wc_regions_clean.shp') # vector (sf package)

## bathymetry raster
depth <- rast('data/depth.tif') # raster (terra package)

## SST rasters (combine SST rasters into a raster stack)
SST8 <- rast('data/average_annual_sst_2008.tif') # raster (terra package)
SST9 <- rast('data/average_annual_sst_2009.tif')
SST10 <- rast('data/average_annual_sst_2010.tif')
SST11 <- rast('data/average_annual_sst_2011.tif')
SST12 <- rast('data/average_annual_sst_2012.tif')

## Merged SST rasters
SST_merge <- c(SST8, SST9, SST10, SST11, SST12)
plot(SST_merge)
```

```
## find the mean SST from 2008-2012 (e.g. create single raster of average SST)
SST_mean <- mean(SST_merge, na.rm = TRUE) # map alegbra

plot(SST_mean)
```

```
## convert average SST from Kelvin to Celsius (hint: subtract by 273.15)
SST_celsius <- SST_mean - 273.15 # map alegbra
```

Process Data

```
# Crop depth raster to match the extent of the SST raster (note: the resolutions
of the SST and depth data do not match)
depth_crop <- crop(depth, SST_celsius)

# Resample depth data
resample_depth <- resample(depth_crop, y = SST_celsius, method = "bilinear")

# Stack the resampled depth and SST data
sst_depth_zones <- c(SST_celsius, resample_depth)
```

Check Resolutions, Extent, and CRS

```
# Resolution
if (all(res(SST_celsius) == res(resample_depth))) {
    print("The resolutions match!")
  } else {
    warning("Update resolutions to match!")
    res(resample_depth) <- res(SST_celsius)
  }
```

```
[1] "The resolutions match!"
```

```
# Extent
  if (ext(SST_celsius) == ext(resample_depth)) {
    print("The extents match!")
  } else {
    warning("Update extents to match!")
    ext(resample_depth) <- ext(SST_celsius)
  }
```

```
[1] "The extents match!"
```

```
# CRS
  if (crs(depth) == crs(SST_celsius)) {
    print("The coordinate reference systems match!")
  } else {
    warning("Update coordinate reference systems to match!")
    project(SST_celsius, depth)
    if (crs(resample_depth) == crs(SST_celsius)) {
      print("The coordinate reference systems now match!")}
  }
```

```
[1] "The coordinate reference systems match!"
```

Update rasters to reflect suitable and not suitable locations for Oysters

```
# Reclassify SST raster cells to suitable(1) and not suitable(0) categories
for oysters
rcl_sst <- matrix(
    c(-Inf, 11, 0, # Not suitable
      11, 30, 1, # Suitable
      30, Inf, 0 # Not suitable
    ),
    ncol = 3,
    byrow = TRUE)

# Reclassify depth raster cells to suitable(1) and not suitable(0) categories
for oysters
rcl_depth <- matrix(
    c(-Inf, -70, 0, # Not suitable
      -70, 0, 1, # Suitable
      0, Inf, 0 # Not suitable
    ),
    ncol = 3,
    byrow = TRUE)

# Classify rasters based on new RCL
zone_SST <- classify(SST_celsius, rcl = rcl_sst) # function produces SST and
depth rastsers with 1 and 0 based on suitability determined by RCL

zone_depth <- classify(resample_depth, rcl = rcl_depth)

# Stacked reclassified rasters
depth_SST_zones <- c(zone_SST, zone_depth)
```

Determine suitable zones based on suitable temperatures and depth

```
# lapp() multiplying 0 and 1 to find suitable sst and depth
suitable_zones <- lapp(depth_SST_zones, fun = function(sst, depth) sst * depth)

# Compare west_cost_eez CRS to suitable_zones CRS
sf::st_crs(west_cost_eez) <- terra::crs(suitable_zones)

# Rasterize: converts vector data into a raster based on suitable_zones raster
ezz_raster <- rasterize(west_cost_eez, suitable_zones, field = 'rgn_id')

# Cellsize() to create a new raster on the same grid as suitable_zones
cellsize <- cellSize(suitable_zones, unit = "km", # unit = "km" gives the area
in km²
mask = TRUE) # mask = TRUE makes cells that are NA in suitable_zones (such as
land) NA here too

# Mask()
suitable_zone_mask <- mask(cellsize, suitable_zones, maskvalues = 0) # Changes
where suitable_zones = 0 to NA only keeping where oysters can live

# Zonal() adds up the suitable area within each EEZ region
zone <- zonal(suitable_zone_mask, ezz_raster, fun = "sum", na.rm = TRUE)

# Create a tibble of suitable zones
zone_tibble <- as_tibble(zone) # vector dataframe

# Joining suitable area to EEZ polygons
eez_map <- left_join(west_cost_eez, zone_tibble, by = "rgn_id")
```

Table output of suitable zones for oysters

```
eez_map |>
  st_drop_geometry() |>
  kable(caption = "Suitable oyster aquaculture area by West Coast EEZ",
    booktabs = TRUE)
```

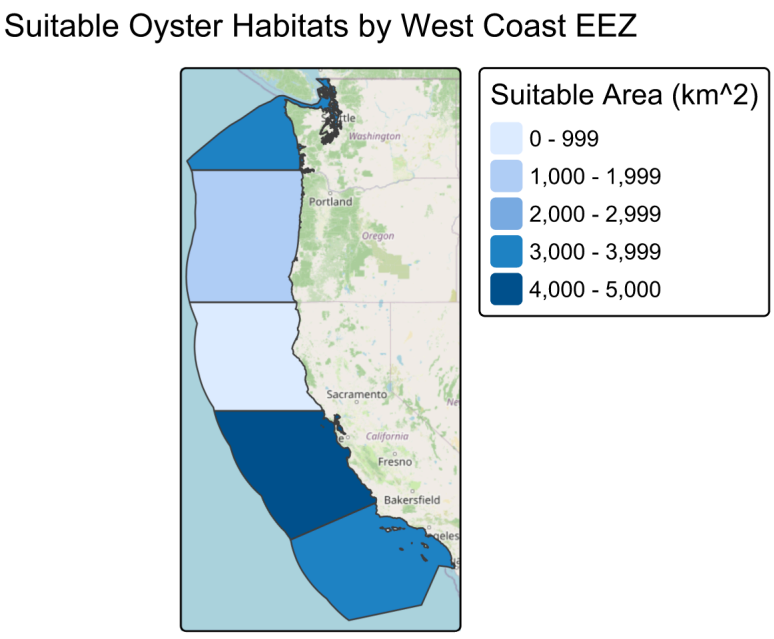
rgn	rgn_key	area_m2	rgn_id	area_km2	area
Oregon	OR	179994061293	1	179994.06	1395.7328
Northern California	CA-N	164378809215	2	164378.81	323.9124
Central California	CA-C	202738329147	3	202738.33	4373.5565
Southern California	CA-S	206860777840	4	206860.78	3329.9002

rgn	rgn_key	area_m2	rgn_id	area_km2	area
Washington	WA	66898309678	5	66898.31	3091.5158

This table highlights each zone's suitable habitat area for oysters. Zone 4, Central California, is the most suitable zone with area of 4373.5565 km² where depth and SST meet optimal conditions.

Map of suitable zones for oysters

```
# Map
tm_shape(eez_map) +
    tm_basemap("OpenStreetMap") +
    tm_fill(
      fill = "area",
      fill.legend = tm_legend(title = "Suitable Area (km^2)")
    ) +
    tmap_options(component.autoscale = FALSE) +
    tm_borders(lwd = .8) +
     tm_title(paste("Suitable", "Oyster", "Habitats by West Coast EEZ"), size
= 1)
```



Function to find suitable_areas for species!

```
suitable_areas    <-    function(minimum_sst,    maximum_sst,    minimum_depth,
maximum_depth, species_name) {
```

```
     # Define a reclassification matrix for temperature
    rcl_sst <- matrix(c(-Inf, minimum_sst, 0, # not suitable
                        minimum_sst, maximum_sst, 1,  # suitable
                        maximum_sst, Inf, 0), # not suitable
                        ncol = 3, byrow = TRUE)

    # Apply the matrix to reclassify the raster, making all cell values 0 (not
suitable) or 1 (suitable)
    sst_rcl <- terra::classify(SST_celsius, rcl = rcl_sst)

    # Define a reclassification matrix for depth
    rcl_depth <- matrix(c(-Inf, minimum_depth, 0, # not suitable
                          minimum_depth, maximum_depth, 1,  # suitable
                          maximum_depth, Inf, 0), # not suitable
                          ncol = 3, byrow = TRUE)

    # Apply the matrix to reclassify the raster, making all cell values 0 (not
suitable) or 1 (suitable)
    depth_rcl <- terra::classify(resample_depth, rcl = rcl_depth)

    # Stacked reclassified rasters of suitable conditions
    combined_rcl <- c(sst_rcl, depth_rcl)

    # lapp() produces new raster by multiplying 0 and 1 to find both suitable
sst and depth loactions
    suitable_zones <- lapp(combined_rcl, fun = function(SST,depth) SST * depth)

   # Rasterize converts vector data into a raster based on suitable_zones raster
    ezz_raster <- rasterize(west_cost_eez, suitable_zones, field = 'rgn_id')

    # Cellsize() to create a new raster on the same grid as suitable_zones
    suitable_cellsize <- cellSize(suitable_zones, unit = "km", mask = TRUE)

    # Mask() changes where suitable_zones = 0 to NA only keeping where species
can live
     suitable_zone_mask <- mask(suitable_cellsize, suitable_zones, maskvalues
= 0)

    # Zonal() adds up the suitable area within each EEZ region
    zone <- zonal(suitable_zone_mask, ezz_raster, fun = "sum", na.rm = TRUE)

    # Create a tibble of suitable zones
    zone_tibble <- as_tibble(zone)
```

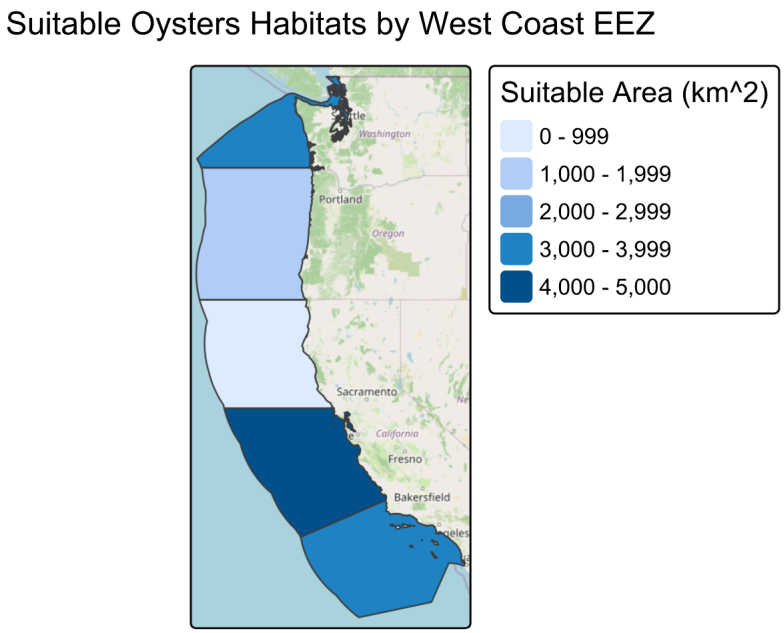
```
    # Joining suitable area to EEZ polygons
    eez_map <- left_join(west_cost_eez,zone_tibble, by= 'rgn_id')

    # Map of suitable zones
    tm_shape(eez_map) +
        tm_basemap("OpenStreetMap") +
        tm_fill(
        fill = "area",
        fill.legend = tm_legend(title = "Suitable Area (km^2)")
        ) +
        tmap_options(component.autoscale = FALSE) +
        tm_borders(lwd = .8) +
       tm_title(paste("Suitable", species_name, "Habitats by West Coast EEZ"),
size = 1)
}
```

Function output for oyster suitable zones

Suitable conditions: - Depth: 0-70 m - Temperature: 11-30 C

```
suitable_areas(minimum_sst = 11, maximum_sst = 30,
            minimum_depth = -70, maximum_depth = 0,
            species_name = "Oysters")
```

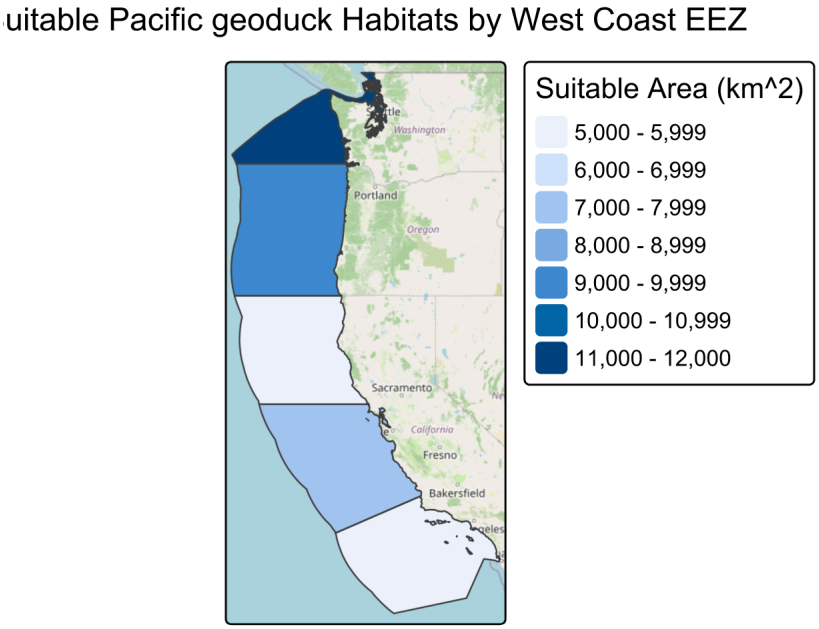


The map highlights that the suitable oyster habitat is concentrated along the California coast, with the largest area extending from around Sacramento southward toward Bakersfield (mainly the central California coast). This area best meets the species' sea surface temperature and depth requirements, making them the most favorable locations for aquaculture development. Additional suitable habitat is found along the Washington coast and parts of Southern California, however these regions are not as suitable. Overall, the results suggest that California's EEZ contains the greatest potential for oyster aquaculture on the U.S. West Coast.

Function output for Pacific geoduck suitable zones

Suitable conditions: - Depth: 0-110 m - Temperature: 7-19 C

```
suitable_areas(minimum_sst = 7, maximum_sst = 19,
               minimum_depth = -110, maximum_depth = 0,
               species_name = "Pacific geoduck")
```



The map highlights that the suitable Pacific geoduck habitat is concentrated along the Washington coast. This area best meets the species' sea surface temperature and depth requirements, making this the most favorable locations for aquaculture development. Additional suitable habitat is found along the Oregon coast. The California coast is the least suitable habitat for the Pacific geoduck population since the water is warmer and outside of their ideal zones. Overall, the results suggest that Washington's EEZ contains the greatest potential for Pacific geoduck aquaculture on the U.S. West Coast.